Aarhus university

Computer-Science

Numerical Linear Algebra

Handin 6

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Math done by a human

Consider the space of differentiable functions $f : [0, \pi] \to \mathbb{R}$ with the L^2 inner product

$$\langle f, g \rangle = \int_0^\pi f(x)g(x)\,dx. \tag{1}$$

Showing Orthogonality

For the given collection of functions defined on $x \in [0, \pi]$

$$\sin(x), \sin(2x), \ldots, \sin(kx), \quad \text{For every integer } k > 0 \tag{2}$$

Using equation(1) and applying the collection(2), it's possible to express the inner product of collection as such:

$$\int_0^\pi \sin(nx)\sin(mx)\,dx \quad \text{for } m \neq n \tag{3}$$

note the $m \neq n$ is used since $\langle f, f \rangle$ is the norm.
This integral can be made simpler by using the trigonometric identity 9.15 [1].

$$\int_0^\pi \frac{1}{2}[\cos((n-m)x) - \cos((n+m)x)]\,dx \quad \text{for } m \neq n$$

then using the integral rules, by taking the constants out and splitting the integral.

$$\frac{1}{2}[\int_0^\pi \cos((n-m)x)\,dx - \int_0^\pi \cos((n+m)x)\,dx] \tag{4}$$

from here each integral can be evaluated, starting with the first one.
Considering the first integral from equation(4), $\cos((n-m)x)$, this can be seen as a composite function on the form $h(i(x))$, where $h(u) = \cos(u)$ and $i(x) = (n-m)x$.
This means the integral can be evaluated by substitution.
Let $u = (n-m)x$ and $du = (n-m)dx$, then express dx as a product of u.

$$du = (n-m)dx$$
$$dx = \frac{1}{(n-m)}du$$

Note the limits $\int_0^\pi$ also change to accomodate u, this is done by putting the value of the limits into $i(x)$, which gives.

$$i(\pi) = (n-m)(\pi) = \pi(n-m)$$
$$i(0) = (n-m)(0) = 0$$

then substitute the new limits into the respective integral that was chosen.

$$\int_0^{\pi} \cos((n-m)x)\,dx \to \int_0^{\pi(n-m)} \cos(u)\frac{1}{(n-m)}\,du$$

remember to take the constant out:

$$\frac{1}{(n-m)}\int_0^{\pi(n-m)} \cos(u)\,du \tag{5}$$

now to integrate and evaluate (5).

$$\begin{aligned}
&\frac{1}{(n-m)}[\sin(u)]_0^{\pi(n-m)} \\
=&\frac{1}{(n-m)}(\sin((n-m)\pi) - \sin(0)) \\
=&\frac{1}{(n-m)}(0-0) \\
=&0
\end{aligned}$$

The reason for this is that $n, m \in k$ (integers) and $\sin(\pi) = 0$. Since any whole multiplier of π is either a half (odd number) or whole (even number) rotation which, in Mr.sin() eyes, is still 0.
It's assumed sin(0) needs no explanation.

With all that in order, the same can be shown for the second integral in equation(4) resulting in:

$$\frac{1}{2}[0-0] = 0$$

which means the inner product of the collection(2) is 0 meaning it's an orthogonal collection.

Constant function $f(x)$

Given the constant funnction $f(x) \equiv 1$, how does that change the orthogonality when k is 'even' or 'odd'?.
Inserting the new $f(x)$ in equation(1).

$$\langle f, g\rangle = \int_0^{\pi} 1 \cdot g(x)\,dx.$$

this prevents the use of trigonominal identities resultig in evaluating cos() instead of sin() in the integral.

By evaluating the integral via substitution, like in the previous integral.

$$\begin{aligned}
&\int_0^{k\pi} \sin(u)\frac{1}{k}\,du \\
&= \frac{1}{k}\int_0^{k\pi} \sin(u)\,du \\
&= \frac{1}{k}[-\cos(u)]_0^{k\pi} \\
&= \frac{1}{k}(-\cos(k\pi) - (-\cos(0)))
\end{aligned}$$

Here is the clutch, $(-\cos(0) = -1)$ and depending on how many π rotations, $(-\cos(k\pi))$ is either 1 or -1. For an odd k $(-\cos(k\pi) = 1)$ and for even $(-\cos(k\pi) = -1)$, which changes the result in step (4) above to either 0 (even) or some non-zero number $w \in [-1, 0)$ (odd)
Note: The author knows the notation could be simplified, but it made more sense to be explicit with the signs in the moment of greatness.

Projection

Determinign the projection onto $\sin(x)$ and $\sin(3x)$ using $f(x) \equiv 1$ yiels:

$$proj_{\sin(x)}(1) = \frac{\int_0^{\pi} 1 \cdot \sin(x)\,dx}{\int_0^{\pi} \sin(x) \cdot \sin(x)\,dx} \sin(x) \tag{6}$$

$$proj_{\sin(3x)}(1) = \frac{\int_0^{\pi} 1 \cdot \sin(3x)\,dx}{\int_0^{\pi} \sin(3x) \cdot \sin(3x)\,dx} \sin(3x) \tag{7}$$

Using the results from the previous section, the numerator in(6) and(7) can be generalized to $\frac{2}{k}$ since both $\sin(x)$ and $\sin(3x)$ use odd k values.
Before inserting anything the denominator shall be evaluated, starting with equation(6).

$$\int_0^{\pi} \sin(x) \cdot \sin(x)\,dx$$

The above integral can be simplified using the trigonometric identities 9.15 [1] but for $\alpha = \beta$ and $\alpha = x$

$$2\sin^2(x) = -\cos(2x) + \cos(0)$$

isolate $\sin^2(x)$ to fit the case, remember $\cos(0) = 1$

$$\sin^2(x) = \frac{1 - \cos(2x)}{2} \tag{8}$$

substituting that into the integral

$$\int_0^{\pi} \frac{1 - \cos(2x)}{2}$$

extracting the constant and evaluating gets the following

$$\frac{1}{2}[x - \sin(2x)]_0^\pi$$

when evaluated

$$\frac{1}{2}((\pi - 0) - (0 - 0)) = \frac{1}{2}\pi = \frac{\pi}{2} \tag{9}$$

It's important to note for this, that the result from integrating equation(8) using either $\sin^2(x)$, as was just done, or $\sin^2(3x)$ would yield the same result $\frac{\pi}{2}$ since $\sin(2x)$ and $\sin(6x)$ both equal 0.
By combining the findings for the denominator(9) and the numerator(see start of section **Projection**), the equations(6)(7) can be rewritten as:

$$proj_{\sin(x)}(1) = \frac{2}{\frac{\pi}{2}}\sin(x) = \frac{4}{\pi}\sin(x) \tag{10}$$

$$proj_{\sin(3x)}(1) = \frac{\frac{2}{3}}{\frac{\pi}{2}}\sin(3x) = \frac{4}{3\pi}\sin(3x) \tag{11}$$

these projections represents (mathematically) the place on the sine waves the looks like the line $f(x) \equiv 1$ the most. According to equation (13.3) [1] adding the projections together will result in a better approximation of line on the span.

The given $g(x) = 1 + c_1\sin(x) + c_3\sin(3x)$ looks a bit like the (c) from proposition 13.10 [1], the residual vector.

$$f(x) - (proj_{\sin(x)}(1) + proj_{\sin(3x)}(1)) \perp w \tag{12}$$

and remember $f(x) \equiv 1$, so the difference is that $g(x)$ adds the projected vector P to the original vector V (see cited proposition for notation) instead of subtracting it, which wouldn't make the resulting vector orthogonal to the defined span $[\sin(x), \sin(3x)]$ (trust).
So to make $g(x)$ orthogonal to the span, the sign has to be cancelled out, this can be done by 'reversing' the constants c_1 and c_3 from equation(10)(11).

$$\frac{3}{\pi}\sin(x) + \frac{4}{3\pi}\sin(3x) \to (-\frac{3}{\pi}\sin(x)) - \frac{3}{3\pi}\sin(3x) \tag{13}$$

this should cancel out the '+' sign in $g(x)$ and make it orthogonal, which can be shown by evaluating the inner product $g(x), \sin(x)$.

1

References

[1] Andrew Swan. *Numerisk Lineær Algebra*. Institut for Matematik, Aarhus Universitet, 2025.